POWER GENERATION AND MONITORING



PRESENTED BY

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INTRODUCTION:

- Rapid energy demand and fossil fuel depletion highlight the need for renewable sources.
- Solar energy is abundant and eco-friendly but requires optimal positioning for efficiency.
- The project develops an automated solar tracking system to maximize solar panel energy capture.
- Uses Light Dependent Resistors (LDRs) and a motorized base for automatic sun alignment.
- Enables real-time monitoring of power output for efficiency and sustainability.

OBJECTIVE:

- To maximize solar power generation through automatic tracking and monitoring.
- To design a low-cost, IoT-enabled solar system for efficient renewable energy usage.
- To minimize human intervention using an automated control mechanism.
- To monitor power output in real-time and improve energy efficiency.
- Enables real-time monitoring of power output for efficiency and sustainability.



PROJECT DEPENDENCE ON EMBEDDED SYSTEMS:

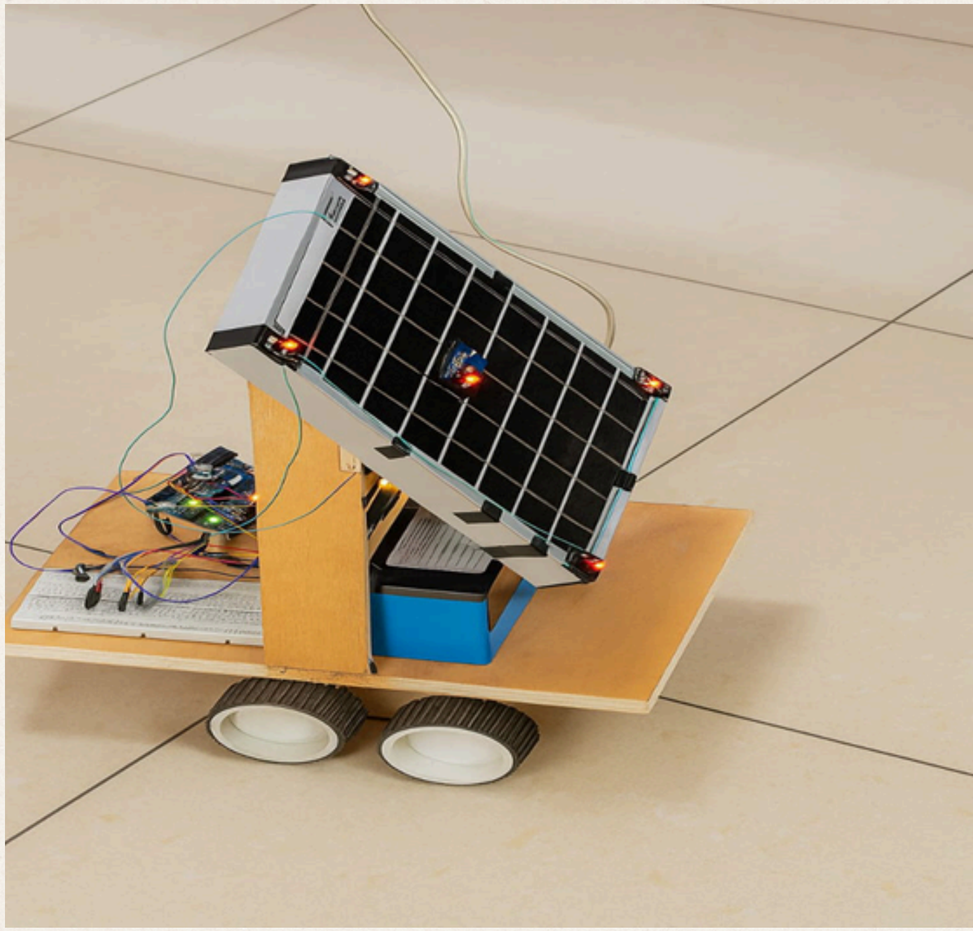
- Uses Arduino Uno microcontroller as the core control unit.
- Embedded system processes sensor data and controls servo/DC motors
- Integration of LDR sensors, motor driver, and IoT modules for smart operation.
- Provides real-time data analysis, control logic, and automation — key features of embedded systems.

COMPONENTS:

Arduino Uno :	Main controller for processing and control
L298N Motor Driver:	Controls rotation of DC motors
Servo Motor:	Adjusts solar panel elevation
LDR Sensors:	Detect light intensity from all directions
Solar Panel:	Converts sunlight into electrical energy
Gear Motors & Wheels:	Enable 360° movement of prototype
IoT Module (optional):	Enables remote monitoring and data logging

WORKING PRINCIPLE:

- Your paraLDR sensors detect sunlight intensity from different directions.
- Arduino processes data to identify the direction of maximum illumination.
- Servo motor adjusts elevation and DC motors adjust azimuth via motor driver.
- The solar panel realigns to maintain maximum exposure automatically.
- IoT system monitors voltage, current, and output parameters remotely.



RESULT:

- The system effectively tracks sunlight and optimizes solar panel orientation.
- Achieved significant improvement in power output compared to fixed panels.
- Automatically adapts to changing sunlight conditions.
- Demonstrated success in real-time monitoring and efficient energy utilization.

WORKING MODEL DISPLAY:

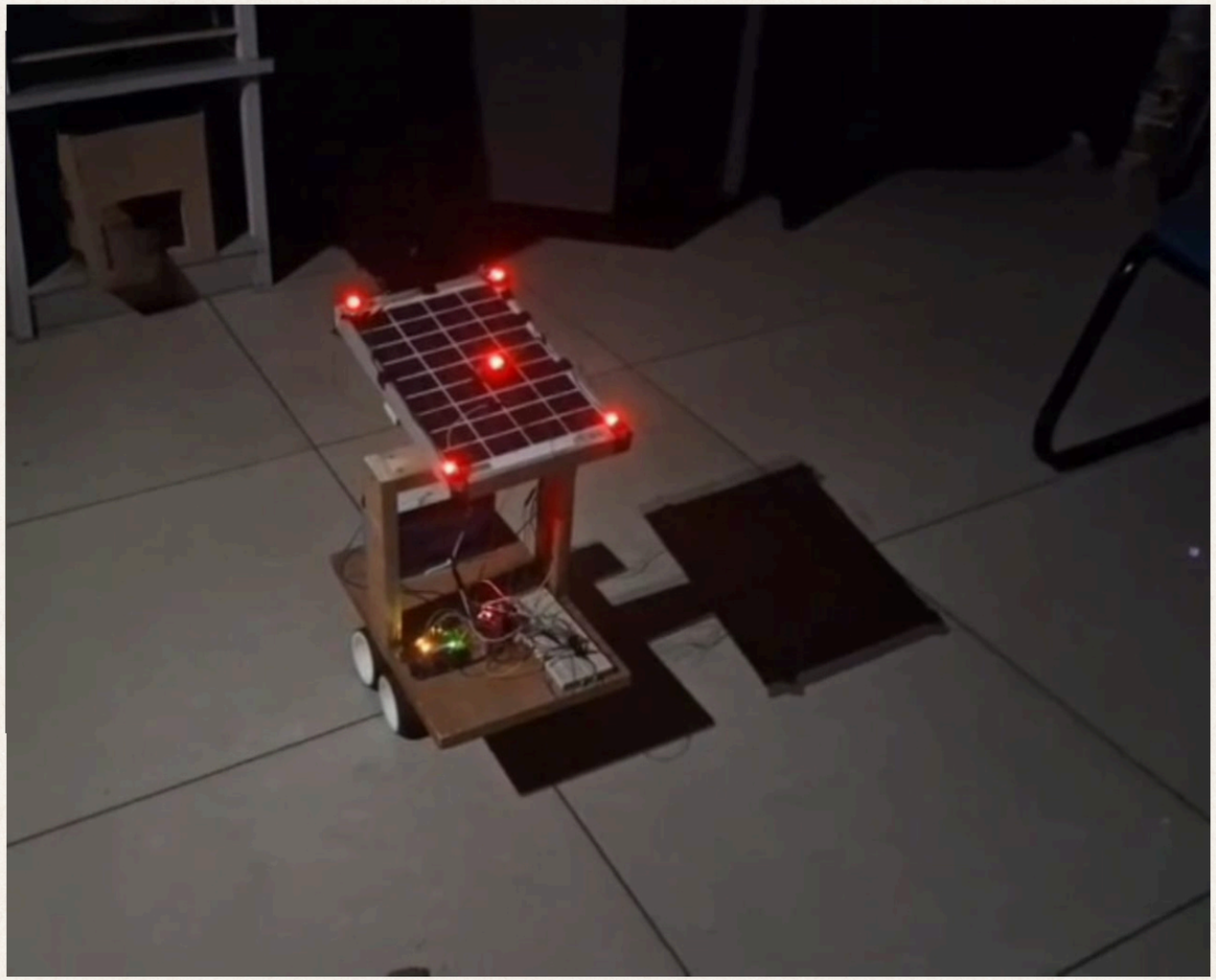


Fig1:Simulation of the project

Arduino Uno

pmfinal.ino

```
1   #include <Servo.h>
2
3   // LDR Pins
```

Serial Monitor ×

Not connected. Select a board and a port to connect automatically.

```
Azimuth: 3 Elevation: 0 Intensity: 642
Azimuth: 3 Elevation: 45 Intensity: 555
Azimuth: 3 Elevation: 60 Intensity: 525
Azimuth: 3 Elevation: 90 Intensity: 499
Azimuth: 4 Elevation: 0 Intensity: 669
Azimuth: 4 Elevation: 45 Intensity: 599
Azimuth: 4 Elevation: 60 Intensity: 526
Azimuth: 4 Elevation: 90 Intensity: 496
Calibration Complete!

Calibration Results:
Azimuth 0: Best Elevation = 60 (Intensity: 548)
Azimuth 1: Best Elevation = 60 (Intensity: 544)
Azimuth 2: Best Elevation = 90 (Intensity: 554)
Azimuth 3: Best Elevation = 90 (Intensity: 499)
Azimuth 4: Best Elevation = 90 (Intensity: 496)

Optimal Sun Position Found:
Azimuth: 4 (288°), Elevation: 90° (Intensity: 496)

Moving to optimal position...
Now at optimal position!
```

Fig2:Arduino code

FUTURE SCOPE:

- Dual-axis tracking for continuous sun alignment throughout the year.
- Integration with IoT and cloud systems for remote monitoring and data analytics.
- Use of machine learning to predict sun movement and enhance tracking accuracy.
- Scalable for large solar farms and domestic installations.
- Promotes sustainable and clean energy generation reducing carbon footprint.

THANK YOU!